Candidate Problem Set with Full Solutions — Steady Currents, Static Magnetic Fields, Inductance

Nine problems (three per section), mid-level with spread

This document collects the nine candidate problems organized into three sections — one from each of three sources (the original five, the course HW PDFs, and Nayfeh & Brussel) — with full worked solutions following each statement. Difficulty within a section scales from ⋆⋆△ (easier mid) to ⋆⋆⋆ (harder mid).

Steady Currents & Resistance

1.1 — Curved-sides resistance of a quarter washer

Source: Cheng P.4-11 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

Refer to the flat conducting quarter-circular washer of Example 4-4: inner radius a, outer radius b, axial thickness h, uniform conductivity σ, angular extent $\pi/2$. Find the resistance between the two *curved* sides ($r = a$ and $r = b$).

Solution. With voltage applied across the inner and outer curved surfaces, those surfaces become equipotentials and by symmetry the current flows *radially*, $\mathbf{J}(r) = \hat{\mathbf{a}}_\rho J_r(r)$, with the total current I (axial integration) the same at every radius.

At radius r ($a < r < b$) the current crosses a cylindrical segment of area

$$A(r) \;=\; \underbrace{\tfrac{\pi}{2}\, r}_{\text{quarter-circle arc length}} \cdot h \;=\; \frac{\pi r h}{2}.$$

Since $I = J_r A(r) = \sigma E_r \cdot (\pi r h/2)$, we have

$$E_r(r) \;=\; \frac{2I}{\pi \sigma h\, r}.$$

The voltage between the curved faces is

$$V_{ab} \;=\; \int_a^b E_r\, dr \;=\; \frac{2I}{\pi\sigma h}\, \ln\frac{b}{a},$$

so

$$\boxed{R \;=\; \frac{V_{ab}}{I} \;=\; \frac{2}{\pi\sigma h}\, \ln\frac{b}{a}.}$$

1.2 — Parallel-plate capacitor with two lossy dielectrics

Source: HW4, problem P.4-8 *Difficulty: ★★△ (mid)*

A d-c voltage V_0 is applied across a parallel-plate capacitor of area S, with two layered lossy dielectrics of thicknesses d_1, d_2, permittivities ϵ_1, ϵ_2, and conductivities σ_1, σ_2. Find (a) $\mathbf{J}$, (b) $\mathbf{E}_1, \mathbf{E}_2$, and (c) the equivalent RC circuit.

Solution. **(a) Current density.** In steady state, $\boldsymbol{\nabla}\cdot\mathbf{J}=0$. With plane geometry and layers stacked along $\hat{\mathbf{a}}_n$, this forces J_n to be continuous across the layer interface — the same $\mathbf{J}$ threads both dielectrics:

$$J \;=\; \sigma_1 E_1 \;=\; \sigma_2 E_2 \quad \text{(same in both layers)}.$$

The two voltage drops must add to V_0:

$$V_0 \;=\; E_1 d_1 + E_2 d_2 \;=\; J\!\left(\frac{d_1}{\sigma_1}+\frac{d_2}{\sigma_2}\right).$$

Solving for J:

$$\boxed{\mathbf{J} \;=\; \hat{\mathbf{a}}_n \,\frac{V_0\,\sigma_1\sigma_2}{\sigma_2 d_1+\sigma_1 d_2}.}$$

(b) Fields. Dividing by each conductivity:

$$\boxed{\mathbf{E}_1 \;=\; \hat{\mathbf{a}}_n \,\frac{V_0\,\sigma_2}{\sigma_2 d_1+\sigma_1 d_2}, \qquad \mathbf{E}_2 \;=\; \hat{\mathbf{a}}_n \,\frac{V_0\,\sigma_1}{\sigma_2 d_1+\sigma_1 d_2}.}$$

(c) Equivalent circuit. Each layer behaves as a leaky capacitor — a resistor in parallel with a capacitor — and the two layers are stacked in series. Per-layer values:

$$R_i \;=\; \frac{d_i}{\sigma_i S}, \qquad C_i \;=\; \frac{\epsilon_i S}{d_i}.$$

$$\boxed{\text{Equivalent circuit: } (R_1 \parallel C_1) \text{ in series with } (R_2 \parallel C_2).}$$

Check: the DC (zero-frequency) resistance is $R_1+R_2=(d_1/\sigma_1+d_2/\sigma_2)/S$, and the open-circuit capacitance from series capacitors is $C_1C_2/(C_1+C_2)$, as expected for layered dielectrics.

1.3 — Cube with position-dependent conductivity

Source: Nayfeh 7.7 (book) *Difficulty: ★★★ (harder mid)*

A conducting cube of side a has conductivity $\sigma(x)=\alpha(a+x)$. Current flows along x from face S ($x=0$) to S' ($x=a$).

Solution. **(a)** In steady state $\boldsymbol{\nabla}\cdot\mathbf{J}=0$ with $\mathbf{J}=\sigma\mathbf{E}=-\sigma\boldsymbol{\nabla}\Phi$. Assuming $\Phi=\Phi(x)$,

$$\boldsymbol{\nabla}\cdot\left[-\sigma(x)\,\boldsymbol{\nabla}\Phi\right] = -\frac{\mathrm{d}}{\mathrm{d}x}\left[\sigma(x)\frac{\mathrm{d}\Phi}{\mathrm{d}x}\right] = 0 \;\Longrightarrow\; \frac{\mathrm{d}}{\mathrm{d}x}\left[(a+x)\frac{\mathrm{d}\Phi}{\mathrm{d}x}\right] = 0. \quad \checkmark$$

(b) Integrate once: $(a+x)\,\mathrm{d}\Phi/\mathrm{d}x = A$ (a constant). Then $\mathrm{d}\Phi/\mathrm{d}x = A/(a+x)$, and integrating,

$$\Phi(x) = A\,\ln(a+x) + B.$$

The potential difference between faces is

$$\Delta\Phi \;=\; \Phi(0)-\Phi(a) \;=\; A\ln a - A\ln(2a) \;=\; -A\,\ln 2.$$

In magnitude (with $A<0$ when the current flows in $+\hat{\mathbf{a}}_x$):

$$|\Delta\Phi| = A\,\ln 2. \quad \checkmark$$

(c) The current density is

$$J_x = -\sigma(x)\frac{\mathrm{d}\Phi}{\mathrm{d}x} = -\alpha(a+x)\cdot\frac{A}{a+x} = -\alpha A.$$

Remarkably, J_x is *uniform* along the cube (the decreasing field in regions of higher conductivity exactly compensates). The total current through face area $S = a^2$ is

$$I = |J_x|\cdot a^2 = \alpha|A|\,a^2.$$

The resistance is therefore

$$\boxed{R \;=\; \frac{|\Delta\Phi|}{I} \;=\; \frac{|A|\ln 2}{\alpha|A|\,a^2} \;=\; \frac{\ln 2}{\alpha\,a^2}.}$$

Static Magnetic Fields

2.1 — B of a finite straight wire

Source: Cheng P.5-3 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

Current I flows in a straight filament from P_1 to P_2. At point P at perpendicular distance r from the wire, with endpoint angles α_1, α_2: (a) prove $\mathbf{B}_P = \hat{\mathbf{a}}_\phi\,\mu I(\sin\alpha_2 - \sin\alpha_1)/(4\pi r)$; (b) recover the infinite-wire limit.

Solution. **(a)** Align the wire along $\hat{\mathbf{a}}_z$ and let z' parameterize position along the wire, with $z' = 0$ at the foot of the perpendicular from P. The Biot–Savart law gives, for a current element $\hat{\mathbf{a}}_z\,dz'$,

$$dB_\phi = \frac{\mu I}{4\pi}\,\frac{r\,dz'}{(r^2+z'^2)^{3/2}}.$$

Substitute $z' = r\tan\alpha$, so $dz' = r\sec^2\alpha\,d\alpha$ and $(r^2+z'^2)^{3/2} = r^3\sec^3\alpha$:

$$dB_\phi = \frac{\mu I}{4\pi r}\cos\alpha\,d\alpha.$$

Integrate from α_1 to α_2:

$$\boxed{\mathbf{B}_P \;=\; \hat{\mathbf{a}}_\phi\,\frac{\mu I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1).}$$

(b) For an infinite wire, $\alpha_1 \to -\pi/2$, $\alpha_2 \to +\pi/2$, so $\sin\alpha_2 - \sin\alpha_1 = 1-(-1) = 2$, yielding

$$\mathbf{B}_P \;=\; \hat{\mathbf{a}}_\phi\,\frac{\mu I}{2\pi r}. \quad \checkmark$$

2.2 — Coaxial cable: B in every region

Source: HW5, problem P.5-6 *Difficulty: ★★△ (mid)*

Infinite coax: inner conductor radius a carries $+I$; outer shell from b to c carries $-I$ (uniformly distributed currents). Find $\mathbf{B}$ in all regions.

Solution. By cylindrical symmetry $\mathbf{B} = \hat{\mathbf{a}}_\phi B_\phi(\rho)$. Apply Ampère's law, $\oint \mathbf{H}\cdot d\mathbf{l} = I_{\text{enc}}$, on a circle of radius ρ. I compute the enclosed current in each region.

Region 1: $0 < \rho < a$ **(inside inner conductor).** The current is uniformly distributed, so $I_{\text{enc}} = I \cdot \rho^2/a^2$:

$$B_\phi \cdot 2\pi\rho = \mu_0 I \frac{\rho^2}{a^2} \implies \mathbf{B} = \hat{\mathbf{a}}_\phi \frac{\mu_0 I \rho}{2\pi a^2}.$$

Region 2: $a < \rho < b$ **(annular gap).** $I_{\text{enc}} = I$:

$$\mathbf{B} = \hat{\mathbf{a}}_\phi \frac{\mu_0 I}{2\pi\rho}.$$

Region 3: $b < \rho < c$ **(inside outer conductor).** The outer shell carries current density $J = -I/[\pi(c^2 - b^2)]$. Enclosed current:

$$I_{\text{enc}} = I + J \cdot \pi(\rho^2 - b^2) = I \frac{c^2 - \rho^2}{c^2 - b^2}.$$

Therefore

$$\mathbf{B} = \hat{\mathbf{a}}_\phi \frac{\mu_0 I}{2\pi\rho} \frac{c^2 - \rho^2}{c^2 - b^2}.$$

Region 4: $\rho > c$ **(exterior).** $I_{\text{enc}} = I - I = 0$:

$$\mathbf{B} = 0.$$

$$\boxed{\mathbf{B}(\rho) = \hat{\mathbf{a}}_\phi \times \begin{cases} \mu_0 I\rho/(2\pi a^2) & 0 < \rho < a, \\ \mu_0 I/(2\pi\rho) & a < \rho < b, \\ \mu_0 I(c^2 - \rho^2)/[2\pi\rho(c^2 - b^2)] & b < \rho < c, \\ 0 & \rho > c. \end{cases}}$$

The profile rises linearly from 0 at $\rho = 0$ to a peak at $\rho = a$, falls as $1/\rho$ through the gap, declines smoothly to zero at $\rho = c$, and vanishes outside. B_ϕ is continuous at $\rho = a$ and $\rho = c$, as expected in the absence of surface currents.

2.3 — Hairpin wire bent around a cylinder

Source: Nayfeh 8.14 (book) *Difficulty: ★★★ (harder mid)*

A hairpin: two anti-parallel semi-infinite straight legs connected by a 180° arc of radius b. Find $\mathbf{B}$ at the cylinder's center (in the plane of the wire).

Solution. Place the cylinder's axis along $\hat{\mathbf{a}}_z$ and work in the $z = 0$ plane. The wire path has three pieces:

- Upper leg: $y = +b$, x from $-\infty$ to 0, current $+\hat{\mathbf{a}}_x$.
- Arc: radius b, center at origin, from $(0, +b)$ around $(+b, 0)$ to $(0, -b)$.
- Lower leg: $y = -b$, x from 0 to $-\infty$, current $-\hat{\mathbf{a}}_x$.

Each piece contributes to $\mathbf{B}(0)$; I add them up.

Upper leg (semi-infinite). Perpendicular distance from origin to the line is b. Using the finite-wire formula of problem 2.1, the foot of the perpendicular is at $(0, b)$ and z' increases in the $+\hat{\mathbf{a}}_x$ direction. The endpoints are $\alpha_1 = -\pi/2$ (at $x = -\infty$) and $\alpha_2 = 0$ (at $x = 0$), so

$$|\mathbf{B}_{\text{up}}| = \frac{\mu_0 I}{4\pi b}(\sin 0 - \sin(-\tfrac{\pi}{2})) = \frac{\mu_0 I}{4\pi b}.$$

Using the right-hand rule (current in $+\hat{\mathbf{a}}_x$, field point below the wire in $-\hat{\mathbf{a}}_y$), $\mathbf{B}_{\text{up}} = -\hat{\mathbf{a}}_z\,\mu_0 I/(4\pi b)$.

Lower leg (semi-infinite). By symmetry (current reversed *and* location reflected across $y = 0$), this contributes the same magnitude, same direction:

$$\mathbf{B}_{\text{low}} = -\hat{\mathbf{a}}_z\,\frac{\mu_0 I}{4\pi b}.$$

Arc (180° of radius b). A full circular loop gives $\mu_0 I/(2b)$ at its center; a half loop gives half of that:

$$|\mathbf{B}_{\text{arc}}| = \frac{\mu_0 I}{4b}.$$

The current traverses the arc clockwise as viewed from $+\hat{\mathbf{a}}_z$, so $\mathbf{B}_{\text{arc}} = -\hat{\mathbf{a}}_z\,\mu_0 I/(4b)$.

Sum. All three contributions point in $-\hat{\mathbf{a}}_z$:

$$\mathbf{B}(0) = -\hat{\mathbf{a}}_z\left[\frac{\mu_0 I}{4\pi b} + \frac{\mu_0 I}{4\pi b} + \frac{\mu_0 I}{4b}\right] = -\hat{\mathbf{a}}_z\,\frac{\mu_0 I}{4\pi b}\,(2+\pi).$$

$$\boxed{|\mathbf{B}(0)| \;=\; \frac{\mu_0 I\,(\pi+2)}{4\pi b}, \qquad \text{in } -\hat{\mathbf{a}}_z.}$$

Inductance

3.1 — Mutual inductance: long wire & equilateral triangle

Source: Cheng P.5-16 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

Long wire and coplanar equilateral triangle of side b; near vertex at perpendicular distance d, opposite side (length b) parallel to the wire.

Solution. For an equilateral triangle of side b, the perpendicular distance from a vertex to the opposite side is $H = \sqrt{3}\,b/2$. Take x to be the perpendicular distance from the wire to a vertical strip; the strip lies at $x \in [d,\, d+H]$. At distance x, $B(x) = \mu_0 I/(2\pi x)$.

The triangle's height at horizontal position x (measured $s = x - d$ from the near vertex) grows linearly from 0 to b:

$$\ell(s) = \frac{2s}{\sqrt{3}}, \qquad 0 \le s \le H.$$

The flux is

$$\Phi = \int_d^{d+H} B(x)\,\ell(x-d)\,dx = \frac{\mu_0 I}{\pi\sqrt{3}}\int_d^{d+H}\frac{x-d}{x}\,dx.$$

Evaluate the integral:

$$\int_d^{d+H}\frac{x-d}{x}\,dx = H - d\,\ln\frac{d+H}{d} = \frac{\sqrt{3}\,b}{2} - d\,\ln\!\left(1+\frac{\sqrt{3}\,b}{2d}\right).$$

Therefore

$$\boxed{M \;=\; \frac{\Phi}{I} \;=\; \frac{\mu_0 b}{2\pi} - \frac{\mu_0\,d}{\pi\sqrt{3}}\,\ln\!\left(1+\frac{\sqrt{3}\,b}{2d}\right).}$$

3.2 — Self-inductance of a toroid (circular cross section)

Source: HW5, problem P.5-15 *Difficulty: ★★△ (mid)*

Toroidal coil: N turns on an air frame, mean radius r_o, circular cross section of radius b. Find L approximately for $b \ll r_o$.

Solution. Apply Ampère's law to a circular loop of radius ρ inside the toroid: $H(2\pi\rho) = NI$, hence

$$B(\rho) = \frac{\mu_0 N I}{2\pi\rho}.$$

For $b \ll r_o$, B is nearly uniform across the cross section and approximately equal to its value at the mean radius:

$$B \approx \frac{\mu_0 N I}{2\pi r_o}.$$

Flux through one turn:

$$\Phi_1 \approx B \cdot \pi b^2 = \frac{\mu_0 N I\, b^2}{2r_o}.$$

The flux linkage threads N turns:

$$\lambda = N\Phi_1 = \frac{\mu_0 N^2 I\, b^2}{2r_o},$$

and the self-inductance is

$$\boxed{L \;\approx\; \frac{\mu_0 N^2 b^2}{2r_o}. \qquad (b \ll r_o)}$$

Exact form (for comparison). Integrating properly across the circular cross section gives $L_{\text{exact}} = \mu_0 N^2\left(r_o - \sqrt{r_o^2 - b^2}\right)$, which Taylor-expands to $\mu_0 N^2 b^2/(2r_o)$ for $b \ll r_o$. ✓

3.3 — Small coil inside a solenoid

Source: Nayfeh 11.2 (book) *Difficulty: ★★★ (harder mid)*

Solenoid: $n = 100$ turns/cm, diameter 3.0 cm. Inside coil: $N_2 = 20$ turns, diameter 2.0 cm, parallel axis. Solenoid current $I(t) = 3t + 2t^2$. Find (a) induced emf in the coil, and (b) induced current at $t = 2\,\text{s}$ if $R = 0.15\ \Omega$.

Solution. (a) Induced emf. Inside the long solenoid the field is uniform, $B = \mu_0 n I(t)$, directed along the axis. It threads every turn of the inner coil identically. The flux linkage of the inner coil is

$$\lambda(t) = N_2 \cdot B \cdot A_2 = N_2\, \mu_0 n\, A_2\, I(t),$$

where $A_2 = \pi(0.01\,\text{m})^2 = \pi \times 10^{-4}\,\text{m}^2$ is the small coil's cross-sectional area. The mutual inductance is

$$M = N_2\, \mu_0 n\, A_2 = 20 \cdot (4\pi \times 10^{-7}) \cdot (10^4) \cdot (\pi \times 10^{-4}) = 80\pi^2 \times 10^{-7}\,\text{H} \approx 7.90 \times 10^{-5}\,\text{H}.$$

Faraday's law:

$$\varepsilon(t) = -\frac{\mathrm{d}\lambda}{\mathrm{d}t} = -M\,\frac{\mathrm{d}I}{\mathrm{d}t} = -M\,(3 + 4t).$$

$$\boxed{|\varepsilon(t)| \;\approx\; 7.90 \times 10^{-5}\,(3 + 4t)\ \text{V}.}$$

(b) Current at $t = 2\,\mathrm{s}$. At $t = 2$, $dI/dt = 3 + 4(2) = 11\,\mathrm{A/s}$, so

$$|\varepsilon(2)| = (7.90 \times 10^{-5}) \cdot 11 \approx 8.69 \times 10^{-4}\,\mathrm{V}.$$

Ohm's law gives

$$i = \frac{|\varepsilon|}{R} = \frac{8.69 \times 10^{-4}}{0.15} \approx 5.79 \times 10^{-3}\,\mathrm{A}.$$

$$\boxed{i(2\,\mathrm{s}) \;\approx\; 5.8\,\mathrm{mA}.}$$

Remark. Note that the solenoid's diameter (3 cm) never enters the answer — the field inside a long solenoid is independent of radius, so only the *inner* coil's area matters. The solenoid's diameter would matter only if we wanted the flux inside the solenoid as a whole, or needed to check that the inner coil (2 cm) actually fits inside (it does).

Summary of results

Problem	**Result**
1.1	$R = \dfrac{2}{\pi\sigma h}\ln\dfrac{b}{a}$
1.2	$J = \dfrac{V_0\sigma_1\sigma_2}{\sigma_2 d_1 + \sigma_1 d_2}$; $(R_1 \| C_1)$ in series with $(R_2 \| C_2)$
1.3	$R = \dfrac{\ln 2}{\alpha a^2}$
2.1	$\mathbf{B}_P = \hat{\mathbf{a}}_\phi \dfrac{\mu I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1)$
2.2	Piecewise, peak $\mu_0 I/(2\pi a)$ at $\rho = a$; zero for $\rho > c$
2.3	$\lvert\mathbf{B}(0)\rvert = \dfrac{\mu_0 I(\pi+2)}{4\pi b}$
3.1	$M = \dfrac{\mu_0 b}{2\pi} - \dfrac{\mu_0 d}{\pi\sqrt{3}}\ln\left(1 + \dfrac{\sqrt{3}\,b}{2d}\right)$
3.2	$L \approx \dfrac{\mu_0 N^2 b^2}{2r_o}$
3.3	$i(2\,\mathrm{s}) \approx 5.8\,\mathrm{mA}$